

Prob. that the procedure on slide 11.9 produces value $z_i = \underline{P(U \leq p_i)} = p_i$

$$X \longrightarrow F_X(x)$$

$$Y = F_X(X) \text{ transformed r.v.s}$$

$$\begin{aligned} P(Y \leq y) &= P(F_X(X) \leq y) \\ &= P(X \leq F_X^{-1}(y)) \end{aligned}$$

$$\begin{aligned} F_Y(y) &= y = F_X(F_X^{-1}(y)) = y \\ &\longrightarrow Y \sim \text{Uniform}(0,1) \end{aligned}$$

$$\text{Let us say } X \sim N(\mu, \sigma^2) \quad F_X(x) = \Phi\left(\frac{x-\mu}{\sigma}\right)$$

$$Y = F_X(X) \sim U[0,1]$$

$$X = F_X^{-1}(Y) \text{ is Gaussian distr.}$$

$$\begin{aligned} &Y \sim \text{Uniform}[0,1] \\ \text{Then } F_X^{-1}(Y) &\sim N(\mu, \sigma^2) \end{aligned}$$

$$Z = F_X^{-1}(Y)$$

$$\begin{aligned} P(Z \leq z) &= P(F_X^{-1}(Y) \leq z) \\ &= P(Y \leq F_X(z)) \\ &= F_X(z) \end{aligned}$$